Quiz 3 Solutions

Markov Chains and Queuing Systems (EE 621), Spring'26

QUESTION 1

See Fig. 1.

There are four states. Let state 0 mean that no machines are down, state 1 that machine 1 is down and 2 is up, state 2 that machine 1 is up and 2 is down, and state 3 that both machines are down. The balance equations are as follows:

$$\begin{aligned}
(\lambda_1 + \lambda_2)P_0 &= \mu_1 P_1 + \mu_2 P_2 \\
(\mu_1 + \lambda_2)P_1 &= \lambda_1 P_0 \\
(\lambda_1 + \mu_2)P_2 &= \lambda_2 P_0 + \mu_1 P_3 \\
\mu_1 P_3 &= \lambda_2 P_1 + \lambda_1 P_2 \\
P_0 + P_1 + P_2 + P_3 &= 1
\end{aligned}$$

The equations are easily solved and the proportion of time machine 2 is down is $P_2 + P_3$.

Fig. 1. The solution for Question 1.

QUESTION 2

The total arrival rates at the three stations can be obtained by solving the following system of equations:

$$\begin{aligned}
\lambda_1 &= 5, \\
\lambda_2 &= 10 + \frac{\lambda_1}{3} + \frac{\lambda_3}{2}, \\
\lambda_3 &= 15 + \frac{\lambda_1}{3} + \lambda_2.
\end{aligned}$$

The solution is $\lambda_1 = 5$, $\lambda_2 = 40$, and $\lambda_3 = \frac{170}{3}$. The service rates of the three stations are $\mu_1 = 10$, $\mu_2 = 50$, and $\mu_3 = 100$.

(a) The average number of customers in the system is

$$L = \frac{\lambda_1}{\mu_1 - \lambda_1} + \frac{\lambda_2}{\mu_2 - \lambda_2} + \frac{\lambda_3}{\mu_3 - \lambda_3} = \frac{82}{13}.$$

(b) By Little's law, the average amount of time a customer spends in the system is

$$\frac{L}{5+10+15} = \frac{41}{195}.$$

QUESTION 3

This is problem 3.27 on p. 247 of "Data Networks" by Bertsekas and Gallager, for which a solutions manual is available at https://web.mit.edu/dimitrib/www/datanets.html.

QUESTION 4

See Fig. 2.

The number N of sojourns in i has a geometric distribution $\mathbb{P}(N = k) = f^{k-1}(1-f)$, $k \geq 1$, for some $f < 1$. The length of each of these sojourns has the exponential distribution with some parameter g_i. By the independence of these lengths, the total time T in state i has moment generating function

$$\mathbb{E}(e^{\theta T}) = \sum_{k=1}^{\infty} f^{k-1}(1-f)\left(\frac{g_i}{g_i - \theta}\right)^k = \frac{g_i(1-f)}{g_i(1-f) - \theta}.$$

The distribution of T is exponential with parameter $g_i(1-f)$.

Fig. 2. The solution for Question 4.

QUESTION 5

(a) To show that the vector process $X(t) = (X_1(t), \ldots, X_m(t))$ is time-reversible, we must show that it satisfies the following equation for all states i and j:

$$P_i q_{ij} = P_j q_{ji}.$$

Let $i = (i_1, \ldots, i_m)$ and $j = (j_1, \ldots, j_m)$ be two states in the product state space. Because the processes are independent, a transition in the vector process occurs when exactly one of the component processes transitions. If more than one component were to change simultaneously, the transition rate would be zero. Suppose process k transitions from i_k to j_k, while all other $X_n(t)$ remain in state i_n. The transition rate for the vector process is

$$q_{ij} = q^{(k)}_{i_k, j_k},$$

where $q^{(k)}_{i_k,j_k}$ is the transition rate of the kth independent CTMC. Since the processes are independent, the stationary distribution of the vector process is the product of the individual stationary distributions:

$$P_i = \prod_{n=1}^{m} P^{(n)}_{i_n},$$

where $P^{(n)}_{i_n}$ is the stationary probability of state i_n for the nth process. Now,

$$\begin{aligned} P_i q_{ij} &= \left(\prod_{n=1}^{m} P^{(n)}_{i_n}\right) q^{(k)}_{i_k j_k} \\ &= \left(\prod_{n\neq k} P^{(n)}_{i_n}\right) \left(P^{(k)}_{i_k} q^{(k)}_{i_k j_k}\right) \\ &= \left(\prod_{n\neq k} P^{(n)}_{i_n}\right) \left(P^{(k)}_{j_k} q^{(k)}_{j_k i_k}\right), \end{aligned}$$

where the last equality follows from the time-reversibility of the kth process. Since $i_n = j_n$ for all $n \neq k$, we can write

$$P_i q_{ij} = \left(\prod_{n\neq k} P^{(n)}_{j_n}\right) \left(P^{(k)}_{j_k} q^{(k)}_{j_k i_k}\right) = P_j q_{ji}.$$

The result follows.

(b) See Fig. 3.

Consider first the system without the restriction that it is shut down when a single component is working. Letting $X_i(t)$, $i = 1, \ldots, n$, equal 1 if component i is working at time t and 0 if it failed, then $\{X_i(t), t \geqslant 0\}$, $i = 1, \ldots, n$, are independent birth and death processes. Because a birth and death process is time reversible, it follows from part (a) that the process $\{(X_1(t), \ldots, X_n(t)), t \geqslant 0\}$ is also time reversible. Now, with

$$P_i(j) = \lim_{t \to \infty} P\{X_i(t) = j\}, \qquad j = 0, 1$$

we have

$$P_i(1) = \frac{\mu_i}{\mu_i + \lambda_i}, \qquad P_i(0) = \frac{\lambda_i}{\mu_i + \lambda_i}$$

Also, with

$$P(j_1, \ldots, j_n) = \lim_{t \to \infty} P\{X_i(t) = j_i, i = 1, \ldots, n\}$$

it follows, by independence, that

$$P(j_1, \ldots, j_n) = \prod_{i=1}^{n} P_i(j_i), \qquad j_i = 0, 1, \; i = 1, \ldots, n$$

Now, note that shutting down the system when only one component is working is equivalent to truncating the preceding unconstrained system to the set consisting of all states except the one having all components down. Therefore, with P_T denoting a probability for the truncated system, we have that

$$P_T(j_1, \ldots, j_n) = \frac{P(j_1, \ldots, j_n)}{1 - C}, \qquad \sum_{i=1}^{n} j_i > 0$$

where

$$C = P(0, \ldots, 0) = \prod_{j=1}^{n} \lambda_j/(\mu_j + \lambda_j)$$

Hence, letting $(0, 1_i) = (0, \ldots, 0, 1, 0, \ldots, 0)$ be the n vector of zeroes and ones whose single 1 is in the ith place, we have

$$\begin{aligned} P_T(\text{system is shut down}) &= \sum_{i=1}^{n} P_T(0, 1_i) \\ &= \frac{1}{1 - C} \sum_{i=1}^{n} \left(\frac{\mu_i}{\mu_i + \lambda_i}\right) \prod_{j \neq i} \left(\frac{\lambda_j}{\mu_j + \lambda_j}\right) \\ &= \frac{C \sum_{i=1}^{n} \mu_i/\lambda_i}{1 - C} \end{aligned}$$

Let R denote the number of components being repaired. Then with I_i equal to 1 if component i is being repaired and 0 otherwise, we have for the unconstrained (nontruncated) system that

$$E[R] = E\left[\sum_{i=1}^{n} I_i\right] = \sum_{i=1}^{n} P_i(0) = \sum_{i=1}^{n} \lambda_i/(\mu_i + \lambda_i)$$

But, in addition,

$$\begin{aligned} E[R] &= E[R|\text{all components are in repair}]C \\ &\quad + E[R|\text{not all components are in repair}](1 - C) \\ &= nC + E_T[R](1 - C) \end{aligned}$$

implying that

$$E_T[R] = \frac{\sum_{i=1}^{n} \lambda_i/(\mu_i + \lambda_i) - nC}{1 - C} \qquad \blacksquare$$

Fig. 3. The solution for Question 5(b).